3 September 1964

SUPERSEDING
MIL-G-10049 (Ord)
28 NOVEMBER 1949

MILITARY SPECIFICATION

CANNON, 76MM GUN: M32

1. SCOPE

1.1 This specification covers one type of gun cannon.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, Mortars, Recoilless Rifles and Guns, General Specification for

MIL-M-11472 - Magnetic Particle Inspection Process for Ferromagnetic Material

DRAWINGS

U. S. Army Weapons Command

EPL 7309241 - Engineering Parts List for Cannon, 76MM Gun: M32

PUBLICATIONS

U. S. Army Weapons Command

LSQAP 7309241 - List of Supplementary Quality Assurance Provisions for Cannon, 76MM Gun: M32

PDS P7309241 - Packaging Data Sheet for Cannon, 76MM Gun: M32

LPDS 7309241 - List of Packaging Data Sheets for Cannon, 76MM Gun: M32

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1015

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3. REQUIREMENTS

3.1 Materials and construction. Cannons and parts thereof shall conform to this specification, the drawings and specifications listed in EPL 7309241 and MIL-C-13931.

3.1.1 Bore and chamber surfaces. The bore and chamber surfaces shall be free from material defects such as cracks, laps, pipes, nonmetallic inclusions, slag pockets and machining defects such as tool gouges, tears, burrs, and chatter marks.

3.1.2 Recoil surface. The recoil surface of the tube shall be chrome plated in accordance with the requirements of MIL-C-13931 for chrome plating of the chamber and bore.

3.2 Breech mechanism. The breech mechanism shall operate without evidence of interference, erratic movement or malfunction. Camming of the operating-crank during counter-recoil or manipulation of the operating-handle to open the breech shall rotate the breechblock-crank rearward and downward thereby dropping the breechblock to open the breech. The extractors shall operate in unison to eject the empty case and shall latch the breechblock in the open position. The breech operating mechanism shall return the breechblock to the closed position when the extractors are tripped or when the breech is closed manually. Counter-clockwise rotation of the adjustor shall increase the preload on the closing spring.

3.3 Firing mechanism group. The cocking, percussion and trigger mechanisms shall function without binding, evidence of interference or erratic movement. During descent of the breechblock, the cocking-lever forced by a camming surface on the breech-ring, shall engage the retractor which in turn shall force the guide and firing-pin rearward, compressing the firing-spring and allowing the sear to latch the guide and firing-pin in the cocked position. Rearward movement of the trigger plunger shall pivot the trigger, allowing the sear to rotate and release the percussion mechanism. The compressed firing-spring shall force the firing-pin to protrude beyond the face of the breechblock and thereby compress the retracting spring which shall then withdraw the firing pin behind the front face of the breechblock.

3.3.1 Firing pin protrusion. The firing pin protrusion, measured from the front face of the breechblock, shall be not less than 0.10 nor more than 0.16 inch.

3.3.2 Firing mechanism overtravel. Overtravel of the firing mechanism in cocking shall be not less than 0.05 inch.

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3.4 Proof testing.

3.4.1 First of production. The first cannon and the first repair tube manufactured under a contract shall be subjected to the production engineering tests specified by the responsible commodity arsenal and the firing tests specified in 4.3.

3.4.2 Regular production. The cannons and repair tubes after the first of production shall be capable of withstanding the proof firing tests specified in 4.3.

3.4.3 Material soundness. After the firing tests, all parts shall be free from cracks, fractures, tears, and other injurious defects as evidenced by visual examination and magnetic particle or liquid penetrant testing.

3.4.4 Bore enlargement. Bore diameters across lands after proof firing shall not exceed the basic (minimum) diameter specified on the applicable drawings by more than the amounts shown in Table I.

Table I

Allowable bore enlargement

Distance forward of commencement of rifling	0.1 Inch	1.0 Inch	1 Cal.	2 Cals. to Muzzle
Amount (Inch)	0.006	0.006	0.005	0.004

NOTE: Interpolate to determine the allowable bore enlargement at distances other than the distances listed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection. Inspection shall be performed in accordance with MIL-C-13931 except as otherwise modified or specified herein.

4.2.1 AQL values. The acceptable quality levels are set forth in the SQAPS listed in LSQAP 7309241.

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4.2.2 Recoil surfaces.

4.2.2.1 Before plating. Visually and dimensionally inspect the recoil surface of each tube.

4.2.2.2 After plating. Visually and dimensionally inspect the recoil surface after plating and the post-plating thermal treatment of each tube.

4.2.3 Functioning of breech mechanism. Manually operate all parts of the breech mechanism of each assembled cannon a minimum of two complete cycles. Any evidence of binding, scraping, rubbing, looseness or other malfunction of any component or failure to completely eject an empty case shall be cause for rejection.

4.2.4 Firing mechanism overtravel. Determine overtravel of the firing mechanism by the following procedures: Replace the firing mechanism retainer with a retainer having a hole of proper size and location as to permit a depth gage to reach the rear face of the firing-pin-guide. Lower the breechblock slowly to the point where the sear engages the firing-pin-guide, as indicated by an audible click. Holding the breechblock in this position, measure the distance from the rear face of the breechblock to the rear face of the firing-pin-guide with a depth gage. Holding the depth gage in position against the breechblock, slowly lower the breechblock to the bottom of its travel. Movement of the breechblock will cause the firing-pin-guide to move backward and thereby cause the depth gage reading to change. At some point, the depth gage reading will be a minimum. This point may or may not be at the bottom of the breechblock travel, due to the curved cam surface of the lever plus tolerance build-up on the components involved. Record the minimum depth gage reading. Subtract the second reading from the first to determine the overtravel of the firing mechanism.

4.2.5 Firing pin protrusion. Measure the firing-pin protrusion before assembly of the breechblock into the breech recess. After assembly of the firing mechanism into the breechblock, push the firing pin forward fully and measure the protrusion by use of gage blocks or a depth gage.

4.2.6 Headspace. Headspace is the distance from the rear surface of the tube to the front face of the breechblock. With the breechblock and the tube assembled and the breechblock in the raised position, measure the headspace with gage blocks, or if available, a go-no-go gage. The measurement is taken by lowering the blocks or gage into the space between the rear face of the tube and the front face of the breechblock.

4.3 Proving ground inspection. Cannons and repair tubes supplied under this specification shall be subjected to proving ground inspection, in accordance with the applicable inspection plan at a test agency designated by USATECOM. The inspection shall be done by Government personnel.

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4.3.1 Proof-firing procedure. The first cannon and the first repair tube manufactured under any contract shall be proof fired in accordance with Table II. Subsequent cannons and repair tubes shall be fired in accordance with Table III.

Table II

Proof firing schedule for first of production

Number of Rounds	Percent of Rated Maximum Pressure
1	50
1	75
1	100
2	115
1	100
2	115
1	100

Table III

Proof firing schedule for regular production

Number of Rounds	Percent of Rated Maximum Pressure
1	115

4.3.2 Combined proof firing. The proof firing for acceptance of the cannon may be combined with that of the mount and recoil mechanism. In such combined tests, rounds in addition to those required by Table III may be fired. (See 4.3.4.1)

4.3.3 Firing rate. Proof firing shall be conducted at a rate of fire not to exceed one round per minute.

4.3.4 Inspection after proof firing.

4.3.4.1 Measure the bore diameter for compliance with 3.3.4. When the schedule of Table III is exceeded, not more than eight (8) additional rounds shall be fired prior to bore measurement and examination with a borescope.

4.3.4.2 Exterior surface. Inspect the outer surfaces of the tube, breech-block, breech ring, muzzle brake and evacuator chamber by magnetic particle test conducted in accordance with MIL-M-11472 for compliance with soundness requirements. Acceptance criteria shall be as specified on the applicable drawing.

4.3.4.3 Proof marking. Proof acceptance marking shall be located on the tube near the serial number.

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4.4 Inspection of the preparation for delivery.

4.4.1 The classification of defects and AQL's for examination of the preparations for delivery are as follows:

Categories	Defect	AQL
Major		
101	Illegible or incorrect marking.	1.0
102	Improper level of packaging and packing. (See procurement documents.)	1.0
103	Inadequate cleaning and drying.	1.5
104	Improper preservative application.	1.5
105	Missing or improper barrier material.	1.5
106	Improper or inadequate blocking and bracing.	1.5
107	Improper closure and strapping of containers.	1.5
Minor	Workmanship of container and packaging.	4.0

5. PREPARATION FOR DELIVERY

5.1 Cannons. Cannons shall be prepared for delivery in accordance with PDS P7309241.

5.2 Repair parts. Repair parts shall be prepared for delivery in accordance with the Packaging Data Sheets listed on LPDS 7309241

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Serial numbers for cannons and repair tubes.

(c) List of repair parts required.

(d) Production engineering tests on first of production cannon or repair tube.

(e) Availability of inspection equipment from Ordnance stocks.

(f) Extent of suppliers' responsibility for Government-furnished and supplier-acquired final inspection equipment.

(g) Levels of preservation, packaging and packing.
NOTE: Level "B" is intended to provide adequate and economical protection for multiple domestic shipments, handling, and indoor storage periods up to six months from date of initial packaging. (See 6.2.)

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(h) When a pilot pack is required.

(i) That packages opened for acceptance inspection shall be repackaged at the supplier's expense.

6.2 Storage surveillance. Items preserved and packaged in accordance with level B requirements must be inspected to determine their condition when not used within a six-month period. Items not used within this period should be represerved and repackaged in accordance with this specification as determined necessary by storage conditions and anticipated storage time.

6.3 The responsible arsenal for this cannon is Watervliet Arsenal, Watervliet, New York, 12189.

Custodian:
Army-WC

Preparing activity:
Army-WC (Wvt)

Project No. 1015-A147

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